1

If G is chordal then it has no induced cycles greater than size 3. By performing an edge contraction we are essentially merging two vertices i, j together into a new vertex v that has edges to $N(i) \cup N(j)$.

Suppose G_0 was not chordal, then there exists an induced cycle in G_0 that is greater than size 3. We know that there must be a v in this cycle that is the result of a contraction from G because otherwise it would imply that all vertices are original to G thus G would have an induced cycle greater than 4 meaning G wouldn't be chordal.

Well since v is part of the induced cycle then we know that $N(v)$ only has two nodes in the cycle as well, meaning $N(i) \cup N(j)$ have two nodes in the cycle so we can break into 4 cases:

1. $N(i)$ contains two nodes in the cycle and $N(j)$ has none (note equivalent for $N(j)$)
2. $N(i)$ contains two nodes in the cycle and $N(j)$ has one of the two as well
3. $N(i)$ contains two nodes in the cycle and $N(j)$ has two as well
4. $N(i)$ and $N(j)$ each have one node in the cycle.

For **4.**, this cannot be the case because we know edge i, j exists so then we would have an induced cycle of at least length 5 so G would not be chordal so incorrect.

For **3.** then we could construct our induced cycle as the same 4 nodes but use i or j instead of v so G would not be chordal so again a contradiction

For **2.** and **1.** same as **3.** but substitute i for v

Therefore G_0 must also be chordal because if it is not we showed that G is not chordal creating a contradiction.

2

The three operations are contracting edges, removing loops, removing duplicates

Edge contractions

Suppose G is planar and H is one edge contraction of G show that H is also planar

This one is actually quite fun to show as like an animation because we would draw G and then animate the contraction to show that our resulting H would be planar just by keeping all the other vertices in the same place and moving the two vertices that are being contracted together. As a bit of a more formal proof, if G is planar then when we contract any edge ther would not be any new edges and in G the edges could be drawn in a way that they do not cross over eachother so after the contraction these edges would be able to roughly be drawn in the same

orientation as in G. Suppose an edge in H crosses another edge, then when we uncontract the edge to convert back to G, that cross edge needed to have been incident t our contracted edge (because all others do not move). If the incident crossing edge crosses over a non incident edge then so would that edge in G, if the incident edge crosses another incident edge then it means that the edge crossed an edge from v which would also be impossible because then they would be crossed in G as well.

Therefore edge contractions do not break planarness of graph

Removing Loops

Removing loops is just removing an edge i, i. If G was already planar then H would be planar because we did not add an edge and only removed a face

Removing duplicate edges

Removing duplicate edges same thing as loops, we were planar in G and we did not add edges and instead removed an edge so leaving all vertices as they are would result in H also being planar

3

Base case is K_2 + a vertex which is 3 colorable (or 2 colorable).

Suppose I add a have a graph with k vertices that is 3 colorable and suppose I add a new vertex with some edges, I need to show that $G + v$ is 3 colorable xor contains a K_4 minor.

Proof by contrapositive is if $G + v$ is not 3-colorable then $G + v$ contains a K_4 minor. v has to be the fourth color because G is three colorable, thus v must have an edge to all of the other colors $1, 2, 3$ which we can name v_1, v_2, v_3. Suppose there is no path from v_i to v_j that does not use v, then we can cycle all the colors in that component until $v_i = v_j$ then $G + v$ would be three colorable, which creates a contradiction because we are supposing that $G + v$ is not 3 colorable. Thus since there exists a path from each of the vertices to eachother, we are able to contract edges along those paths such that we have our K_4 minor.

Therefore if G does not contain a K_4 minor then G is 3-colorable